DPP No: 04

Maximum Time 50 Min

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C³

(COMPETISHUN CRASH COURSE)

PHYSICS

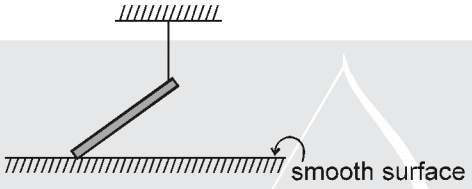
TARGET JEE-MAINS

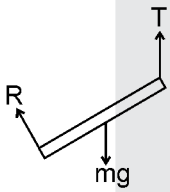
SYLLABUS : NEWTONS LAWS OF MOTION , FRICTION

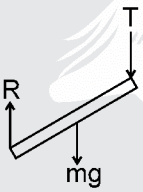
1. Let E,G and N represents the magnitude of electromagnetic, gravitational and nuclear forces between two protons at a given separation (1 fermi). Then

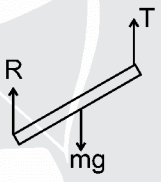
(A) N < E < G (B) E > N > G (C) G > N > E (D) N > E > G

2. Which figure represents the correct F.B.D. of rod of mass m as shown in figure :



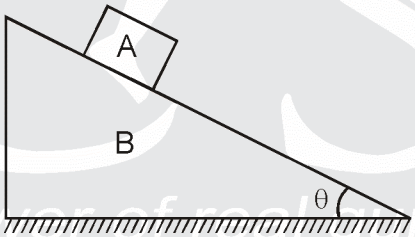
(A) 

(B) 

(C) 

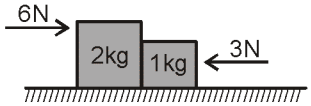
(D) None of these

3. A block is placed on a smooth wedge of inclination angle θ with the horizontal. What minimum acceleration should be given to the wedge in the horizontal direction so that the block starts moving up along the incline?



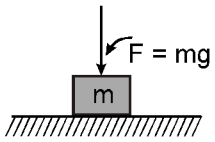
(A) > g cot θ (B) > g tan θ (C) > g sin θ (D) > g cos θ

4. Two forces of 6N and 3N are acting on the two blocks of 2kg and 1kg kept on frictionless floor. What is the force exerted on 2kg block by 1kg block ?



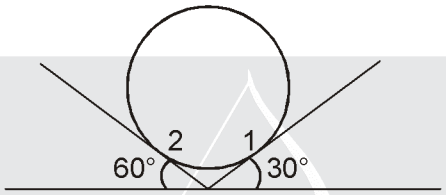
(A) 1N (B) 2N (C) 4N (D) 5N

5. A block of mass 'm' is placed on ground and an additional force F = mg is applied on the block as shown in figure. Calculate contact force between ground and block.



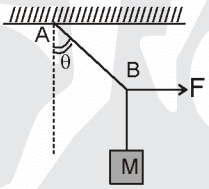
(A) 1 mg (B) 2 mg (C) 3 mg (D) 4 mg

6. A solid sphere of mass 10 kg is placed over two smooth inclined planes as shown in figure. The normal reactions at 2 is 10x N. Find x ($g = 10\ m/s^2$)



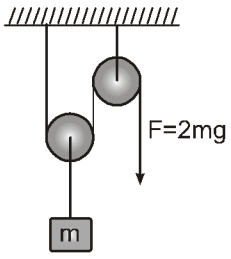
(A) 1 (B) 3 (C) 5 (D) 12

7. A mass M is suspended by a rope from a rigid support at A as shown in figure. Another rope is tied at the end B, and it is pulled horizontally with a force F. If the rope AB makes an angle θ with the vertical in equilibrium, then the tension in the string AB is :



(A) F sin θ (B) F/sin θ (C) F cos θ (D) F/cos θ

8. In the shown mass pulley system, pulleys and string are massless. The one end of the string is pulled by the force F = 2mg. The acceleration of the block will be



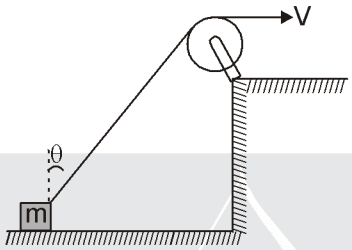
(A) g/2 (B) 0 (C) g (D) 3g

9. A particle of small mass m is joined to a very heavy body by a light string passing over a light pulley. Both bodies are free to move. The total downward force on the pulley due to string is nearly

(A) mg (B) 2 mg

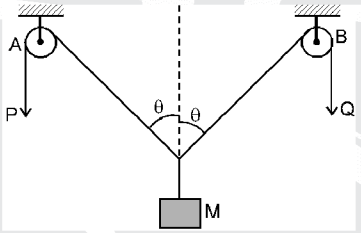
(C) 4 mg (D) can not be determined

10. A block is dragged on smooth plane with the help of a rope which moves with velocity v. The horizontal velocity of the block is :



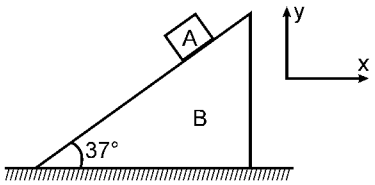
(A) v (B) $\frac{v}{\sin\theta}$ (C) v sin θ (D) $\frac{v}{\cos\theta}$

11. In the arrangement shown in fig. the ends P and Q of an unstretchable string move downwards with uniform speed U. Pulleys A and B are fixed. Mass M moves upwards with a speed.



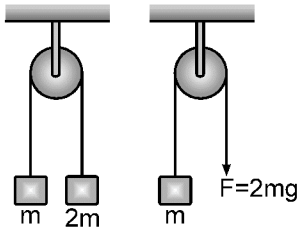
(A) 2 U cos θ (B) U cos θ (C) 2 U/cos θ (D) U/cos θ

12. In the figure shown the acceleration of A is, $\vec{a}_A = 15\hat{i} + 15\hat{j}$ then the acceleration of B is: (A remains in contact with B)



(A) $6\hat{i}$ (B) $-15\hat{i}$ (C) $-10\hat{i}$ (D) $-5\hat{i}$

13. Figure shows two pulley arrangements for lifting a mass m. In **(a)** the mass is lifted by attaching a mass 2 m while in **(b)** the mass is lifted by pulling the other end with a downward force F = 2 mg, If f_a & f_b are the accelerations of the two masses then (Assume string is massless and pulley is ideal)



(A) $f_a = f_b$ (B) $f_a = f_b/2$ (C) $f_a = f_b/3$ (D) $f_a = 2f_b$

14. A particle is moving with a constant speed along a straight line path. A force is not required to :

(A) increase its speed (B) decrease its momentum

(C) change the direction (D) keep it moving with uniform velocity

15. An object will continue accelerating until :

(A) resultant force on it begins to decrease

(B) its velocity changes direction

(C) the resultant force on it is zero

(D) the resultant force is at right angles to its direction of motion

16. A block of mass M is pulled along a smooth horizontally surface with a rope of mass m by force F. The acceleration of the block will be

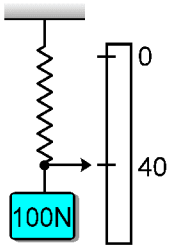
(A) $\frac{F}{(M+m)}$ (B) $\frac{F}{(M-m)}$ (C) $\frac{F}{M}$ (D) $\frac{F}{m}$

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17. A fireman wants to slide down a rope. The rope can bear a tension of $\frac{3}{4}^{th}$ of the weight of the man. With what minimum acceleration should the fireman slide down :

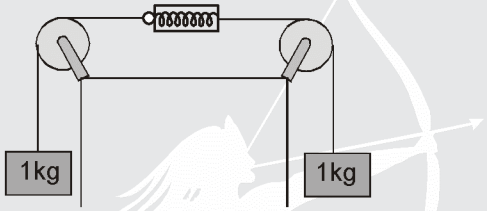
(A) $\frac{g}{3}$ (B) $\frac{g}{6}$ (C) $\frac{g}{4}$ (D) $\frac{g}{2}$

18. An ideal spring, with a pointer attached to its end, hangs next to a scale. With a 100 N weight attached and in equilibrium, the pointer indicates '40' on the scale as shown. Using a 200 N weight instead results in '60 on the scale. Using an unknown weight 'X' instead results in '30' on the scale. The value of 'X' is :



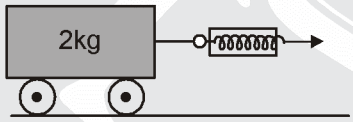
(A) 80 N (B) 60 N (C) 50 N (D) 40 N

19. In the given figure, what is the reading of the spring balance ?



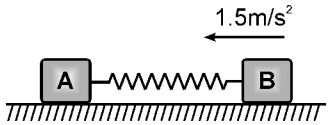
(A) 10 N (B) 20 N (C) 5 N (D) zero

20. A massless spring balance is attached to 2 kg trolley and is used to pull the trolley along a flat surface as shown in the fig. The reading on the spring balance remains at 10 kg during the motion. The acceleration of the trolley is (Use g = 9.8 ms^{-2})



(A) 4.9 ms^{-2} (B) 9.8 ms^{-2} (C) 49 ms^{-2} (D) 98 ms^{-2}

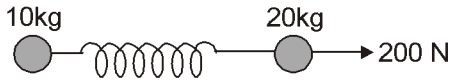
21. Two blocks A & B with mass 4 kg and 6 kg respectively are connected by a stretched spring of negligible mass as in figure. When the two blocks are released simultaneously the initial acceleration of B is 1.5 m/s^2 westward. The acceleration of A is:



(A) 1 m/s^2 westward (B) 2.25 m/s^2 eastward

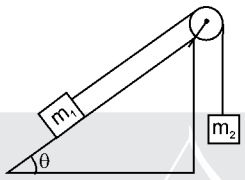
(C) 1 m/s^2 eastward (D) 2.75 m/s^2 westward

22. Two masses of 10 kg and 20 kg respectively are connected by a massless spring as shown in figure. A force of 200 N acts on the 20 kg mass at the instant when the 10 kg mass has an acceleration of 12 ms^{-2} towards right, the acceleration of the 20 kg mass is :



(A) 2 ms^{-2} (B) 4ms^{-2} (C) 10ms^{-2} (D) 20ms^{-2}

23. Two masses m_1 and m_2 are attached to the ends of a massless string which passes over a frictionless pulley attached to the top of an inclined plane. The angle of inclination of the plane in θ. Take g = 10 ms^{-2}



If m_1 = 10kg, m_2 = 5kg, θ = 30º, what is the acceleration of mass m_2?

(A) zero (B) (2/3) ms^{-2} (C) 5 ms^{-2} (D) 10/3 ms^{-2}

24. A lift is falling with an acceleration 2 m/s^2. A ball of mass 100 gm is attached at one end of the string and the other end is fixed to the ceiling of the lift. The ball remains at rest relative to lift. The tension in the string is: (g = 10 m/s^2)

(A) 1.2 N (B) 0.8 N (C) 10 N (D) 0.2 N

25. An object of mass 2 kg moving with constant velocity $10\hat{i}$ m/s is seen in a frame moving with constant velocity $10\hat{i}$ m/s. What will be the value of 'pseudo force' acting on object in this frame.

(A) F = 1 (B) F = 0 (C) F = 2 (D) 2F = 6

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ANSWER KEY

1.	(D)	**2.**	(C)	**3.**	(B)	**4.**	(C)	**5.**	(B)
6.	(C)	**7.**	(B)	**8.**	(D)	**9.**	(C)	**10.**	(B)
11.	(D)	**12.**	(D)	**13.**	(C)	**14.**	(D)	**15.**	(C)
16.	(A)	**17.**	(C)	**18.**	(C)	**19.**	(A)	**20.**	(C)
21.	(B)	**22.**	(B)	**23.**	(A)	**24.**	(B)	**25.**	(B)